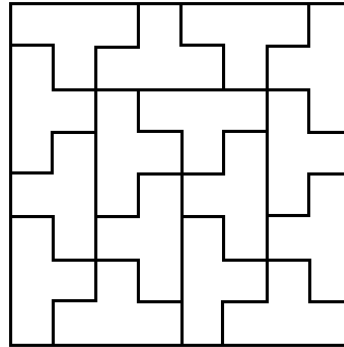
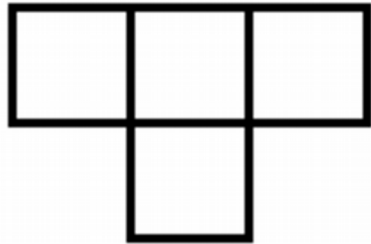


Problem E: Easy Tiling Problem

Time Limit: 5 seconds

Description

Given an $N \times M$ rectangle, compute the number of tilings (complete coverings) with the following piece with 4 blocks (on the left):



Note that the piece can be rotated and flipped but not cut. An example tiling of an 8×8 rectangle is given above right.

Input

A number of of inputs (≤ 100), each line with N and M ($4 \leq N \leq 24$, $4 \leq M \leq 10^9$). Additionally, we stipulate the condition that both N and M are integer multiples of 4 (i.e. $4|N$ and $4|M$).

Output

For each input, output the answer modulo **1000000007** on one line.

Sample Input

```
4 4
4 8
```

Sample Output

```
2
6
```